

Survey of Extreme Learning Machine (ELM) Papers for Computational PDEs

Compiled by ChatGPT

May 1, 2025

1 Igel'nik & Pao (1995): Stochastic Choice of Basis Functions and the Functional-Link Net

Algorithmic Idea

The paper introduces the *random vector functional-link* (RVFL) network, a single-hidden-layer model in which the inner-layer parameters $(\mathbf{w}_k, b_k)$ are *sampled once* from a user-chosen distribution and *kept fixed*. Only the output weights α_k are trained. Function approximation is interpreted as Monte-Carlo quadrature of an integral representation

$$f(\mathbf{x}) = \int_V T[f](\lambda) G_\lambda(\mathbf{x}) d\lambda \approx \frac{|V|}{n} \sum_{k=1}^n T[f](\lambda_k) G_{\lambda_k}(\mathbf{x}), \quad (1)$$

so the expected L_2 error decays like $\mathcal{O}(n^{-1/2})$ independently of input dimension.

Theory

- RVFL is a universal approximator for $C(K)$ on bounded $K \subset \mathbb{R}^d$.
- Convergence rate in mean-square norm is $C/\sqrt{n}$ with C depending on the variance of the integrand but *not* on d .

Numerical Evidence

Experiments on synthetic functions (e.g. Poisson kernel on $[-\pi, \pi]$) confirm the $n^{-1/2}$ decay and show orders-of-magnitude speed-ups versus gradient descent MLPs.

2 Huang et al. (2006): Extreme Learning Machine—Theory and Applications

Algorithm

1. Draw $\{\mathbf{w}_k, b_k\}_{k=1}^N$ i.i.d. from a continuous distribution.
2. Compute hidden matrix $H_{jk} = g(\mathbf{w}_k^\top \mathbf{x}_j + b_k)$.
3. Solve $H\beta = T$ in the least-squares sense via Moore-Penrose pseudo-inverse: $\beta = H^\dagger T$.

Key Theoretical Results

- Any SLFN with infinitely differentiable activation reaches *zero training error* with at most N hidden nodes for N samples.
- Among all LS solutions ELM finds the minimum-norm output weights, favouring generalisation.

Benchmarks

Across 28 UCI tasks, ELM trains $\approx 10^3$ – 10^5 times faster than BP while matching or exceeding generalisation accuracy; on the 10 000-sample MNIST subset it reaches 2.09% test error in 2.1 s.

3 Liu et al. (2015): Is Extreme Learning Machine Feasible? (Part I)

Contributions

- Provides *generalisation bounds*: with polynomial or sigmoid activations, ELM attains the optimal $m^{-2r/(2r+d)}$ rate (Sobolev smoothness r).
- Estimates the required hidden width $n = \mathcal{O}(m^{d/(2r+d)} \log m)$ to reach that rate.
- Proves full-rank of the hidden matrix for polynomial activations, guaranteeing pseudo-inverse solvability.

Numerical Validation

Simulations on synthetic regression with varying d corroborate the predicted rates and confirm the hidden matrix rank behaviour.

4 Dwivedi & Srinivasan (2020): Physics-Informed Extreme Learning Machine (PIELM)

Algorithm

PIELM minimises the composite loss $\mathcal{L} = \|\mathcal{N}[u_\theta]\|_{2,\Omega_{\text{int}}}^2 + \|u_\theta - g\|_{2,\partial\Omega}^2$ where $\mathcal{N}$ is the PDE residual. Hidden parameters are random (ELM) so training reduces to a *linear* LS problem for the output weights. The authors also propose DPIELM—domain-partitioned PIELM—for large domains.

Findings

- For Poisson and advection–diffusion ($d = 3$ – 10) PIELM matches PINN accuracy while being $\approx 1000\times$ faster.
- Exponential error decay w.r.t. hidden width until saturation.

5 Dwivedi & Srinivasan (2020): Biharmonic Equation in Complicated Geometries with PIELM

Highlights

Extends PIELM to fourth-order PDEs with mixed boundary data. On irregular 2-D domains (star-shaped, crescent) the method achieves $< 10^{-5}$ L^∞ error using $\lesssim 1$ s training time—orders faster than PINN and competitive with adaptive FEM without meshing overhead.

6 Li et al. (2022): ELM with Kernels for Elliptic PDEs

Method

Replaces random feature map by *kernel ELM* (KELM); the solution is expressed $u(x) = \sum_{i=1}^N \alpha_i K(x, x_i)$ with $\alpha = (H + \lambda I)^{-1} T$. Hyper-parameters (regularisation λ , kernel width) selected by grid search.

Experiments

On 3 test Poisson-type problems KELM outperforms wavelet NN and butterfly-optimised WNN. Mean absolute error drops below 10^{-4} with only $N = 200$ interior points.

7 Dong & Li (2021): Local Extreme Learning Machines and Domain Decomposition

Algorithm

1. Partition domain into sub-domains; on each use a shallow ELM with M random neurons.
2. Impose C^k continuity across interfaces.
3. Form global LS (linear or nonlinear) for output weights; solve via block time-marching for unsteady PDEs.

Key Observations

Error decays nearly exponentially with M or sub-domain count. For Burgers and Navier–Stokes examples locELM attains 10^{-7} error and beats FEM and PINN in both accuracy and CPU (up to $100\times$).

8 Dong & Yang (2021): Optimising the Random-Coefficient Magnitude R_m

Contribution

Uses differential evolution to minimise the linear-system residual $\|H\beta - T\|_2$ over R_m (single or per-layer). Multi- R_m improves accuracy by up to one order and cuts training time $> 50\%$.

9 Ni & Dong (2022): Hidden-Layer Concatenated ELM (HLConcELM)

Idea

Expose *all* hidden nodes to the output layer via logical concatenation; this guarantees non-decreasing representational capacity as layers widen/stack. HLConcELM retains linear training yet achieves high accuracy even when the final hidden layer is narrow.

Results

For Poisson, Helmholtz and Burgers equations HLConcELM with narrow last layer achieves the same 10^{-8} accuracy as conventional wide-layer ELM but with 60% fewer parameters.

10 Dong & Wang (2023): Inverse Parametric PDEs with Random-Weight NNs

Algorithms

Three strategies:

1. **NLLSQ**: joint optimisation of inverse parameters and output weights via perturbed Gauss–Newton LS.
2. **VarPro-F1**: eliminate inverse params, solve for NN weights by LS, then invert linear system for parameters.
3. **VarPro-F2**: eliminate NN weights, solve reduced problem for inverse params, optional Newton loop for nonlinear PDEs.

Performance

On 1-D inverse advection and 2-D Poisson coefficient-recovery problems, parameter error drops exponentially with collocation points; noise-robust variants maintain $< 2\%$ error at 5% Gaussian noise—significantly better than PINN baselines.